

Classifying simple modules at minimal model central charges through symmetric polynomials

Simon Wood

The Australian National University

Joint work with David Ridout

Outline

- ▶ Review of vertex operator algebras
- ▶ Virasoro minimal models
- ▶ Symmetric polynomials
- ▶ Classifying simple minimal model modules

Motivation

Bottom up approach to constructing conformal field theories:

- ▶ Pick a vertex operator algebra
- ▶ Classify modules in a suitable category
- ▶ Compute characters and their modular properties
- ▶ Find modular invariants
- ▶ Compute fusion products of modules
- ▶ Compute correlation functions

Definition: Vertex operator algebra

A VOA $(V, |0\rangle, T, Y)$ is a collection of data:

- ▶ $\mathbb{N}_0$ -graded vector space $V = \bigoplus_{n \geq 0} V[n]$
- ▶ vacuum vector $|0\rangle \in V[0]$
- ▶ conformal vector $T \in V[2]$
- ▶ linear map (called field map)

$$Y : V \rightarrow \text{End}(V)[[z, z^{-1}]],$$

$$A \mapsto Y(A; z),$$

$$Y(A; z) = \sum_n A_n z^{-n-h_A}, \quad A \in V[h_A]$$

Definition: Vertex operator algebra

The VOA data satisfy the axioms:

- ▶ Compatibility with grading: For $A \in V[h_A]$, $A_n : V[m] \rightarrow V[m - n]$
- ▶ Vacuum and conformal vectors: $Y(|0\rangle; z) = 1$,
 $Y(T; z) = T(z) = \sum_n L_n z^{-n-2}$

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$$

- ▶ $V[n]$ is eigenspace of L_0 with eigenvalue n .
 L_{-1} is the translation operator
 $[L_{-1}, Y(A; z)] = Y(L_{-1}A; z) = \partial_z Y(A; z)$
- ▶ Locality: $\exists N \in \mathbb{N}_0$ such that

$$(z - w)^N [Y(A; z), Y(B; w)] = 0 \text{ in } \text{End}(V)[[z, z^{-1}, w, w^{-1}]]$$

Definition: Vertex operator algebra

Prop

For $A, B, C \in V$

$$\left. \begin{aligned} Y(A; z)Y(B; w)C &\in V((z))((w)) \\ Y(B; w)Y(A; z)C &\in V((w))((z)) \\ Y(Y(A; z - w)B; w)C &\in V((w))((z - w)) \end{aligned} \right\} \begin{array}{l} \text{element of} \\ V[[z, w]][z^{-1}, w^{-1}, (z - w)^{-1}] \\ \text{expanded in 3 domains} \end{array}$$

We identify

$$Y(A; z)Y(B; w) \simeq Y(Y(A; z - w)B; w) \simeq Y(B; w)Y(A; z)$$

and call it the operator product expansion (OPE) of $Y(A; z)$ and $Y(B; w)$.

Example: The universal Virasoro VOA Vir_c

- ▶ $\text{Vir}_c = U(\bigoplus_{n \geq 2} \mathbb{C}L_{-n})|0\rangle$, central charge $c \in \mathbb{C}$,
 $L_n|0\rangle = 0, n \geq -1$
- ▶ vacuum $|0\rangle$, conformal vector $T = L_{-2}|0\rangle$
- ▶ $Y(L_{-2}|0\rangle; z) = T(z)$,
- ▶ $Y(L_{-n_1-2} \cdots L_{-n_i-2}|0\rangle; z) = : \left(\frac{\partial^{n_1}}{n_1!} T(z) \right) \cdots \left(\frac{\partial^{n_i}}{n_i!} T(z) \right) :$
- ▶ Operator product expansion:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg}$$

Example: The Heisenberg VOA

Heisenberg Lie algebra: $\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}a_n \oplus \mathbb{C}\mathbf{1}$, $[a_m, a_n] = m\delta_{m+n,0}\mathbf{1}$

Heisenberg VOA:

- ▶ $\mathcal{H} = U(\bigoplus_{n \geq 1} \mathbb{C}a_{-n})|0\rangle = \mathbb{C}[a_{-1}, a_{-2}, \dots]|0\rangle$, $a_n|0\rangle = 0, n \geq 0$
- ▶ vacuum: $|0\rangle$
- ▶ conformal vector: $T = \frac{1}{2}(a_{-1}^2 + \beta a_{-2})|0\rangle$, $c = 1 - 3\beta^2$
- ▶ $Y(a_{-1}|0\rangle; z) = a(z) = \sum_n a_n z^{-n-1}$,
 $Y(a_{-n_1-1} \cdots a_{-n_i-1}|0\rangle; z) = : \left(\frac{\partial^{n_1}}{n_1!} a(z) \right) \cdots \left(\frac{\partial^{n_i}}{n_i!} a(z) \right) :$
- ▶ Operator product expansion:

$$a(z)a(w) = \frac{1}{(z-w)^2} + \text{reg}$$

Prop

$\text{Vir}_c \rightarrow \mathcal{H}$, $T(z) \mapsto \frac{1}{2} : a(z)^2 : + \frac{\beta}{2} \partial a(z)$,

where $c = 1 - 3\beta^2$, defines an embedding of VOAs. This embedding is called a **free field realisation**.

Definition: VOA modules

- ▶ $\mathbb{C}$ -graded vector space $W = \bigoplus_{h \in \mathbb{C}} W[h]$, $W[h]$ generalised L_0 eigenspace, eigenvalue h .
- ▶ $Y_W : V \rightarrow \text{End}(W)[[z, z^{-1}]]$ such that for $A, B \in V$ and $C \in W$

$$\left. \begin{aligned} Y_W(A; z)Y_W(B; w)C &\in W((z))((w)) \\ Y_W(B; w)Y_W(A; z)C &\in W((w))((z)) \\ Y_W(Y(A; z-w)B; w)C &\in W((w))((z-w)) \end{aligned} \right\} \begin{array}{l} \text{element of} \\ W[[z, w]][z^{-1}, w^{-1}, (z-w)^{-1}] \\ \text{expanded in 3 domains} \end{array}$$

VOA module examples

- ▶ V is a V -module.
- ▶ A proper subspace $I \subset V$ that is a V -module is an ideal of V :
 $Y(V; z)I \subset I((z)), Y(I; z)V \subset I((z))$. If $|0\rangle, T \notin I$, then V/I is a VOA.
- ▶ For Vir_c , Virasoro Verma modules $V_h = U(\bigoplus_{n \geq 1} \mathbb{C}L_{-n})|h\rangle$,
 $L_0|h\rangle = h|h\rangle, L_n|h\rangle = 0, n \geq 1$.
- ▶ For $\mathcal{H}$, Heisenberg Verma modules $F_\lambda = U(\bigoplus_{n \geq 1} \mathbb{C}a_{-n})|\lambda\rangle$,
 $a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$.

Ideals of the universal Virasoro VOA

Prop

The universal Virasoro VOA Vir_c has a proper non-trivial ideal if and only if

$$c = 1 - 6 \frac{(p - q)^2}{pq}, \quad p, q \geq 2, \quad \gcd\{p, q\} = 1.$$

This ideal I is unique and is generated by a singular vector of conformal weight $(p - 1)(q - 1)$.

- ▶ The simple quotient $M(p, q) = \text{Vir}_c / I$ is called the minimal model VOA.
- ▶ A Vir_c -module W is a $M(p, q)$ -module if and only if I acts trivially: $Y(A; z)W = 0, \forall A \in I$.
- ▶ Today's goal: Classify $M(p, q)$ -modules from Vir_c -module theory.

Example: Yang-Lee model

- ▶ central charge $c = -\frac{22}{5}$ Virasoro VOA ($p = 2, q = 5$)
- ▶ maximal proper ideal generated by a singular vector χ

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle$$

$$Y(\chi; z) = :T(z)^2: - \frac{3}{10}\partial^2 T(z) = \sum_{n \in \mathbb{Z}} \chi_n z^{-n-4}$$

- ▶ Evaluate χ_0 on a candidate Virasoro highest weight vector $|h\rangle$.

$$0 = \chi_0|h\rangle = (L_0^2 + \frac{1}{5}L_0)|h\rangle = h(h + \frac{1}{5})|h\rangle$$

- ▶ Conformal weight can only be 0 or $-\frac{1}{5}$.

Classifying minimal model modules

- ▶ Previous example was not a fluke. Up to some “technicalities” (codified in Zhu’s algebra), VOA module theory is completely characterised by the action of zero modes on highest weight vectors.
- ▶ Previous example depended heavily on explicit form of the singular vector. Singular vectors are **very** hard to find in general. No general closed formulae are known.
- ▶ Goal for the rest of the talk: Find a better way to evaluate zero modes of singular vectors on candidate highest weight vectors.

Classifying minimal model modules

- ▶ Previous example was not a fluke. Up to some “technicalities” (codified in Zhu’s algebra), VOA module theory is completely characterised by the action of zero modes on highest weight vectors.
- ▶ Previous example depended heavily on explicit form of the singular vector. Singular vectors are **very** hard to find in general. No general closed formulae are known.
- ▶ Goal for the rest of the talk: Find a better way to evaluate zero modes of singular vectors on candidate highest weight vectors.

Zhu’s algebra:

- ▶ Zhu’s algebra is the associative algebra of zero modes acting on vectors in modules which are annihilated by positive modes.
- ▶ Constructible as a vector space quotient of V .
$$\text{Zhu}(V/I) = \text{Zhu}(V)/\text{Zhu}(I).$$
- ▶ Simple V -modules $\xleftrightarrow{1:1}$ simple Zhu-modules.

Singular vectors in the free field realisation

The Heisenberg VOA $\mathcal{H}$ is special, because one can define a field map for vectors in modules, not just vectors in the VOA.

Define

- ▶ new operator $\hat{a}$: $[a_n, \hat{a}] = \delta_{n,0}$,
- ▶ $e^{\lambda \hat{a}}|\mu\rangle = |\lambda + \mu\rangle$.

Vertex operator (not to be confused with a field in a VOA)

- ▶ $Y(|\lambda\rangle; z) = V_\lambda(z) = e^{\lambda \hat{a}} z^{\lambda a_0} \prod_{m \geq 1} \exp\left(\lambda \frac{a_{-m}}{m} z^m\right) \prod_{m \geq 1} \exp\left(-\lambda \frac{a_m}{m} z^{-m}\right)$
- ▶ $T(w)V_\lambda(z) = \frac{h_\lambda}{(w-z)^2} V_\lambda(z) + \frac{1}{w-z} \partial V_\lambda(z) + \text{reg}$
- ▶ $V_\lambda(z)V_\nu(w) = (z-w)^{\lambda\nu} : V_\lambda(z)V_\nu(w) :$
- ▶ $V_\lambda(z) : F_\mu \rightarrow F_{\lambda+\mu}[[z, z^{-1}]]z^{\lambda\mu}$

Singular vectors in the free field realisation

Screening operator:

- ▶ If $h_\lambda = 1$ then the field $S(z) = V_\lambda(z)$ is called a screening field.

$$T(w)S(z) = \frac{1}{(w-z)^2}S(z) + \frac{1}{w-z}\partial S(z) + \text{reg} = \partial_z \frac{S(z)}{w-z} + \text{reg}$$

- ▶ Zero mode $S_0 = \text{Res}_{z=0} [S(z)]$ is called a screening operator and defines a Virasoro homomorphism:
 $[T(w), S_0] = \text{Res}_{z=w} [T(w)S(z)] = 0$
- ▶ **However**, the map $S_0 : F_\mu \rightarrow F_{\mu+\lambda}$ is only well defined if $\lambda\mu \in \mathbb{Z}$.
- ▶ If $\lambda\mu \in \mathbb{Z}$ and $S_0|\mu\rangle \neq 0$ then $S_0|\mu\rangle$ is a **singular vector** in $F_{\mu+\lambda}$

The singular vector at minimal model central charges

Let $p, q \geq 2$, $\gcd\{p, q\} = 1$ and

$$\alpha_+ = \sqrt{\frac{2q}{p}}, \quad \alpha_- = -\sqrt{\frac{2p}{q}}, \quad \beta = \alpha_+ + \alpha_-, \quad c = 1 - 3\beta^2 = 1 - 6\frac{(p-q)^2}{pq}.$$

There are two screening fields

$$S_+(z) = V_{\alpha_+}(z), \quad S_-(z) = V_{\alpha_-}(z).$$

The singular vector that generates the ideal of $\mathbf{Vir}_c \subset \mathcal{H}$ by which one must quotient to get the minimal model VOA:

$$\chi = S_+^{[p-1]}|(1-p)\alpha_+\rangle \propto S_-^{[q-1]}|(1-q)\alpha_-\rangle \in \mathbf{Vir}_c \subset \mathcal{H}.$$

The constraints on the minimal model conformal weights are obtained by evaluating the action of the zero mode of χ .

Composing screening operators

To construct the singular vector of $\text{Vir}_c \subset \mathcal{H}$, we must consider higher powers of the screening operator.

$$S_+(z_1) \cdots S_+(z_r) = e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ + a_0} \\ \prod_{m \geq 1} \exp \left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m \right) \exp \left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m} \right)$$

Prop

If and only if $\mu = \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-$, $k \in \mathbb{Z}$, there exists a unique choice of a cycle Γ such that

$$S_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-} \rightarrow F_{\frac{1+r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$$

is a non-trivial Virasoro homomorphism.

- ▶ if $k \geq 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle = 0$.
- ▶ if $k < 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle \neq 0$.

Appearance of symmetric polynomials

Action of $S_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$:

$$S_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} \exp\left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m\right) \exp\left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}\right) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^t = \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^t f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This inner product is symmetric and it vanishes if $\deg(f) \neq \deg(g)$.

Symmetric polynomials

One can consider symmetric polynomials in either a finite or infinite number of variables:

- ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$

- ▶ $\Lambda = \varprojlim_r \Lambda_r$

Numerous well known generators:

- ▶ Power sums

$$p_k(z) = \sum_i z_i^k$$

- ▶ Elementary symmetric polynomials

$$e_k(z) = \sum_{1 \leq i_1 < \dots < i_k} z_{i_1} \cdots z_{i_k}$$

$$\Lambda_r = \mathbb{C}[p_1, \dots, p_r] = \mathbb{C}[e_1, \dots, e_r]$$

$$\Lambda = \mathbb{C}[p_1, p_2, \dots] = \mathbb{C}[e_1, e_2, \dots]$$

Symmetric polynomials

Numerous well known bases. Many indexed by partitions

$\lambda = [\lambda_1, \lambda_2, \dots]$.

- ▶ Power sum basis: $p_\lambda = p_{\lambda_1} p_{\lambda_2} \cdots$
- ▶ Elementary symmetric basis: $e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$
- ▶ Monomial symmetric polynomials:

$$m_\lambda(z) = \sum_{\sigma} z_1^{\lambda_{\sigma(1)}} z_2^{\lambda_{\sigma(2)}} \cdots,$$

where the σ are all distinct permutations of $\lambda = [\lambda_1, \lambda_2, \dots]$.

$$m_{[2,2]}(z_1, z_2) = z_1^2 z_2^2, \quad m_{[2,2]}(z_1, z_2, z_3) = z_1^2 z_2^2 + z_1^2 z_3^2 + z_2^2 z_3^2.$$

$$\begin{aligned} \Lambda_r &= \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} p_\lambda = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} e_\lambda = \bigoplus_{\lambda, \ell(\lambda) \leq r} \mathbb{C} m_\lambda \\ \Lambda &= \bigoplus_{\lambda} \mathbb{C} p_\lambda = \bigoplus_{\lambda} \mathbb{C} e_\lambda = \bigoplus_{\lambda} \mathbb{C} m_\lambda \end{aligned}$$

Jack symmetric polynomials

Jack symmetric polynomials $P_\lambda^t(z)$ are uniquely characterised by

1. Upper triangular: $P_\lambda^t(z) = m_\lambda(z) + \sum_{\nu < \lambda} c_{\nu,\lambda}(t)m_\nu(z)$
2. Orthogonal: $\langle P_\lambda^t, P_\nu^t \rangle_r^t = 0$, if $\lambda \neq \nu$.

Well studied with many known expansions:

- ▶ $\langle P_\lambda^t(z), P_\lambda^t(z) \rangle_r^t = a_\lambda(t)$
- ▶ $\prod_{i,j}(1 - y_i z_j)^{-t} = \prod_{m \geq 1} \exp\left(t \frac{p_m(y)p_m(z)}{m}\right) = \sum_\nu b_\nu(t) P_\nu^t(y) P_\nu^t(z)$
- ▶ $\prod_i (1 + z_i)^{Xt} = \sum_\lambda d_\lambda(X, t) P_\lambda^t(z)$

where $a_\lambda(t)$, $b_\lambda(t)$, $c_{\nu,\lambda}(t)$, $d_\lambda(X, t)$ are rational functions in t . Explicit formulae known.

Examples:

- ▶ $P_{[1,\dots,1]}^t(z) = e_{[1,\dots,1]}(z)$
- ▶ $P_{[s,\dots,s]}^t(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s, \ell([s, \dots, s]) = r$

Determining conformal weights from the zero mode

We want to evaluate the zero mode of the singular vector

$$S_+^{[p-1]}|(1-p)\alpha_+\rangle = \int_{\Gamma} S_+(z_1) \cdots S_+(z_{p-1})|(1-p)\alpha_+\rangle dz_1 \cdots dz_{p-1}.$$

The corresponding field is

$$\begin{aligned} S_+^{[p-1]}V_{(1-p)\alpha_+}(w) \\ = \int_{\Gamma} S_+(z_1 + w) \cdots S_+(z_{p-1} + w)V_{(1-p)\alpha_+}(w) dz_1 \cdots dz_{p-1}. \end{aligned}$$

The easiest way to evaluate such a zero mode is to evaluate matrix elements

$$\langle \tau | S_+^{[p-1]}V_{(1-p)\alpha_+}(w) | \tau \rangle,$$

where $\tau \in \mathbb{C}$ and $\langle \tau | a_{-n} = 0, n \geq 1$.

Determining conformal weights from the zero mode

$$\langle \tau | S_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle = \left\langle P_{[(q-1)^{p-1}]}^{\frac{\alpha_+^2}{2}}(z), \prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} \right\rangle_{p-1}^{\alpha_+^2/2}$$

There are known formulae for expanding

$$\prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} = \sum_{\nu} g_{\nu}(\alpha_+ \tau) P_{\nu}^{\frac{\alpha_+^2}{2}}(z) w^{-|\nu|}.$$

Determining conformal weights from the zero mode

$$\langle \tau | S_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle \propto g_{[(q-1)p-1]}(\alpha_+ \tau) \propto \prod_{r,s} (h_\tau - h_{r,s}),$$

where $1 \leq r \leq p-1$, $1 \leq s \leq q-1$, $(r,s) \simeq (p-r, q-s)$ and

$$h_\tau = \frac{\tau}{2}(\tau - \beta), \quad h_{r,s} = \frac{(qs - pr)^2 - (p - q)^2}{4pq}.$$

Prop

The roots of the polynomial

$$f(h_\tau) = \prod_{r,s} (h_\tau - h_{r,s})$$

classify the highest weights of the minimal models.

Applicability of these free field methods

- ▶ We saw today that symmetric polynomials can be used to “quite easily” classify the simple modules of the minimal models.
- ▶ Essentially the same method also works for free field realisations of affine $\mathfrak{sl}_2$.
- ▶ Also appears to work for $N = 1$ Virasoro minimal models and $\mathfrak{osp}(1|2)$ (work in progress).
- ▶ This method was a crucial tool in the classification of the simple modules of certain logarithmic extensions of the minimal models.

The End

Thank you for coming!